

NO.:	G3214-0904	VERSION:	Y	PAGE:	1
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晶圓廠 雙波長聚焦式橢圓繞射儀操作規範

FAB Dual Wavelength Focus Ellipsometer O.I

壹、機台設備 (Machine facilities)

貳、製程目的 (Purpose of process)

參、操作程序 (Operation procedure)

肆、程式內容 (Content of recipe)

伍、自動化操作事項(Automated communication operation items)

陸、操作注意事項 (Operation notes)

柒、例行作業 (Routine operation)

捌、品質控制 (Quality control)

玖、儀器 MSA 檢定 (MSA machine analysis)

拾、機台之 automation 設定

Machine automated communication setting

拾壹、生效與修訂

(Documentation effectiveness and revision)

晶圓廠 雙波長聚焦式橢圓繞射儀操作規範

FAB Dual wavelength focus Ellipsometer operation guideline

壹、機台設備(Machine facilities)

1. 機台設備為 Rudolph Research 製造之 Dual wavelength focus Ellipsometer 。

The machine is the Dual wavelength focus Ellipsometer produced by Rudolph research

2. 機台型號為 FE-ⅢD 。

Machine Model is FE-ⅢD.

3. 機台代號為 EM-4、EM-5、EM-6

4. Machine code is EM-4、EM-5、EM-6

5. 雷射光源為(Laser Beam Source is):

- (1) 氦氖雷射，波長 6328 A 。

He-Ne Laser, wavelength is 6328 A.

- (2) 二極體雷射，波長 7800 A 。

Diode Laser, wavelength is 7800 A.

貳、製程目的(Purpose of process)

量測 SiO₂、Si₃N₄、Poly-Si、Amorphous-Si 等，單層或多層薄膜之厚度 及 折射率(R.I) 。

To measure the thickness and reflectivity of SiO₂, Si₃N₄, Poly-Si and amorphous-Si at single or multiple layer of films.

參、操作程序(Operation procedure)

1. EA 先將機台代號登錄於自動化系統，以便收集參數量測值，其收集參數量測值之方式詳述於“伍、自動化操作事項” 。

For the first, EA should records the machine code in the automation system for convenience in collecting the parameter, refer to "V. automation communication operation items" 。

NO.:	G3214-0904	VERSION:	Y	PAGE:	3
------	------------	----------	---	-------	---

- 將欲量測之晶片放入標準晶舟（藍色PP材質或 Teflon 晶舟），將此晶舟放在晶舟平台上擺放妥當。

Place the wafer to be measured into the standard cassette (blue PP or Teflon cassette).
Place the cassette properly on the cassette stage.

- 在量測選擇畫面上，選擇左方“Process specification”欄位(如圖一)，欲使用的製程名稱之後按 Enter 進入程式選擇畫面，如圖二。

At measurement selection menu, select the “Process specification” key on the left side (as Figure 1), then press Enter to enter the recipe menu as shown in Figure 2.

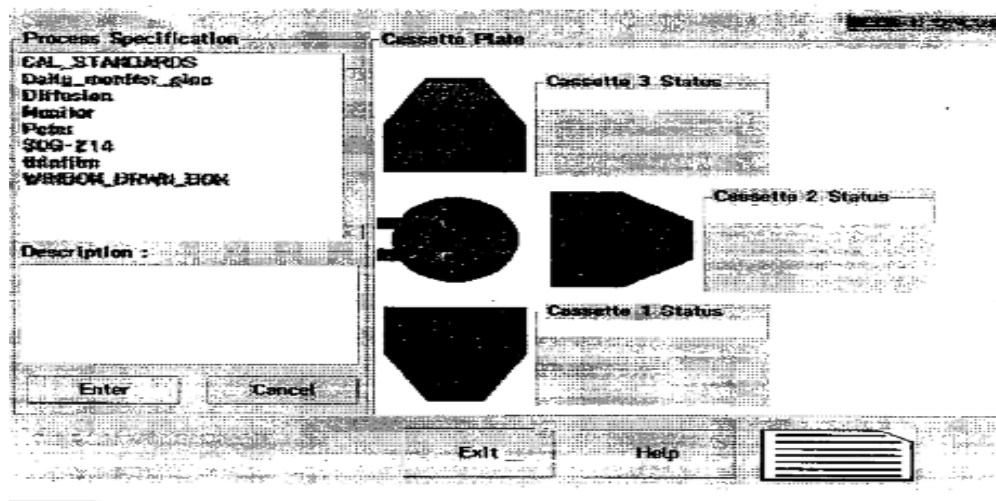


圖 (1)

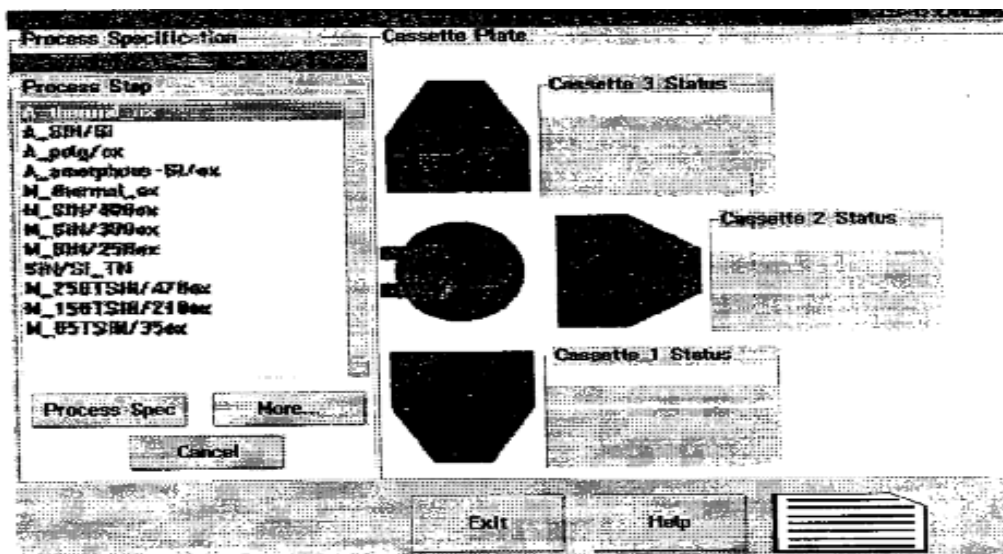


圖 (2)

- 在圖二程式選擇欄“Process Step”中，將箭號移到欲使用的測程式處連續按滑球左邊鍵兩次以選擇該程式，進入選擇晶片位置畫面，如圖三。

Under“ Process Step” in Figure 2, move the arrow to the desired recipe. Double-click the left mouse button to select this recipe to enter the Select Wafer Location Screen as shown in Figure 3.

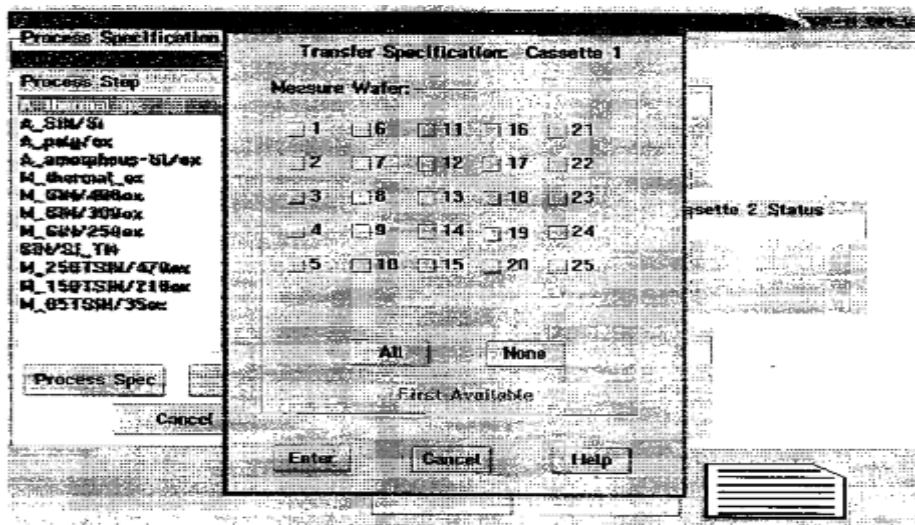


圖 (3)

- 選擇欲量測片在晶舟上之位置後按 Enter 進入預備量測之畫面如圖四，按 Start 進行量測。
Select the location in the cassette to be measured and press Enter the measurement screen as shown in Figure 4 · press start to start measuring.

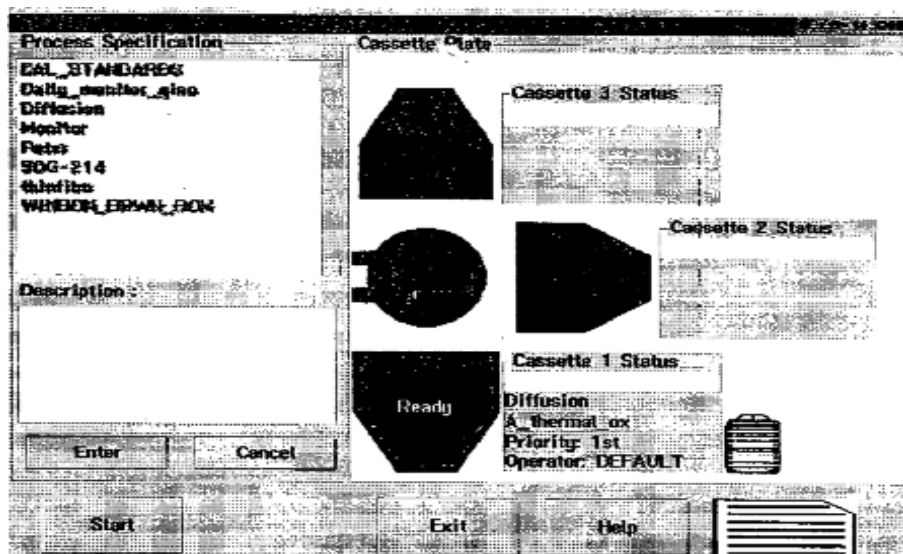


圖 (4)

NO.:	G3214-0904	VERSION:	Y	PAGE:	5
------	------------	----------	---	-------	---

6. 量測完畢回到量測選擇畫面如圖五，在圖一中原晶舟平台位置出現資料紙圖形。移動箭號在料紙上按一下即可顯示出量測資料如圖六。

After measuring, return to the Measure Select Screen as shown in Figure 5. The original cassette station in Figure 1 displays the data sheet pattern. Move the arrow to the data sheet and click to display the measurement data as shown in Figure 6.

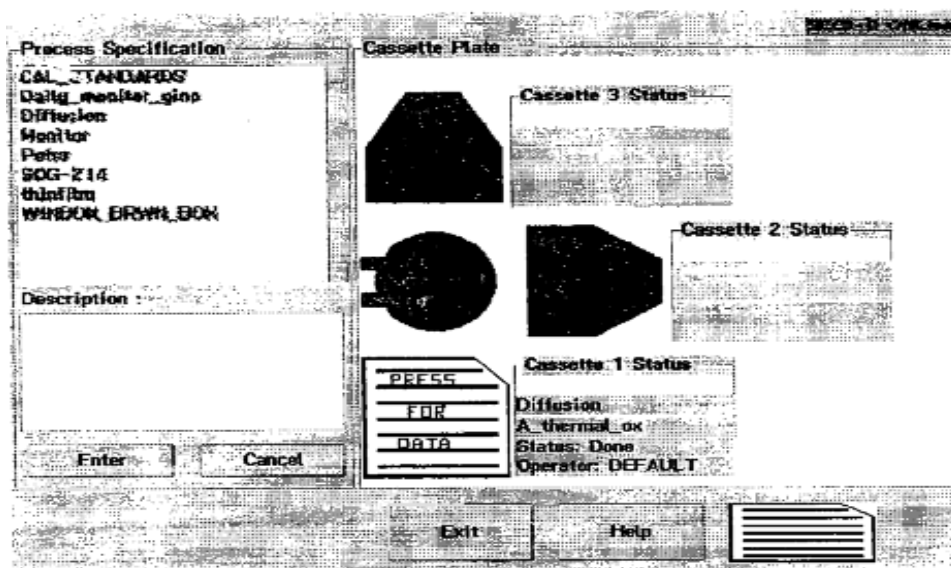


圖 (5)

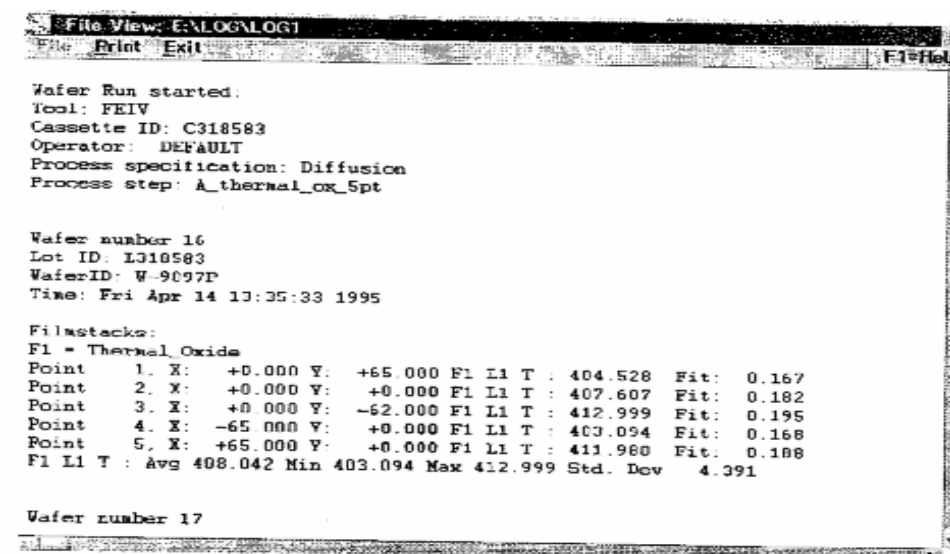


圖 (6)

7. 看完資料後欲離開請移動箭號在 Exit 位置按一下，螢幕上出現詢問資料是否看完之畫面如圖七。選擇Yes 位置按一下離開資料紙同時恢復晶舟平台圖形，以利下一人員量測使用。

NO.:	G3214-0904	VERSION:	Y	PAGE:	6
------	------------	----------	---	-------	---

To exit after reading the data, move the arrow to Exit and click once. The screen asks if you have finished reading the data as shown in Figure 7. Move the arrow to Yes. Click once to exit and restore the cassette station icon for the next measurement.

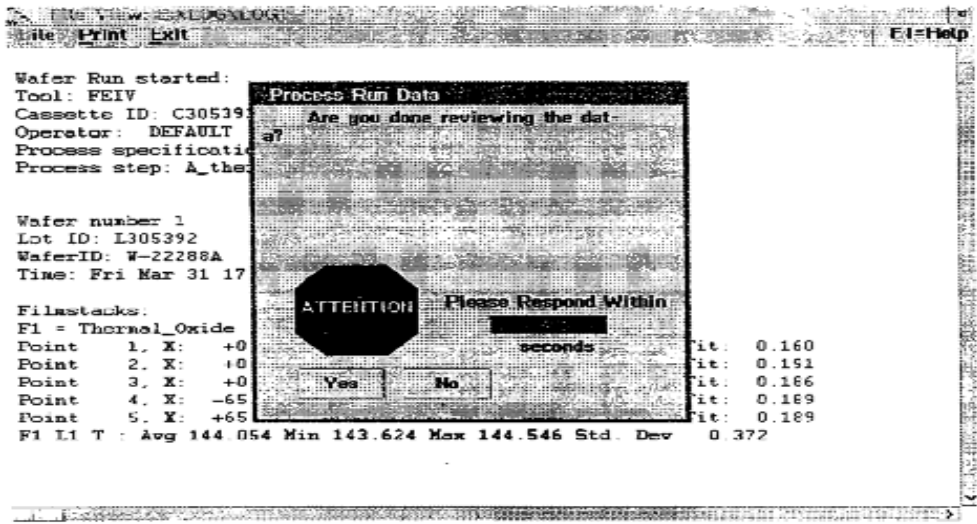


圖 (7)

8. 當第一位人員進行量測時，在量測進行畫面（如圖八）中，將箭號移到 Cassette View 處按一下即可回到量測選擇畫面如圖九。畫面上測平台為紅色表示該平台正在量測中，此時可在有空位量測晶舟平台上按照程序 1~4 之方式進行程式選擇（但不需要再按 Start）即可接著量測。

When the first operator is measuring, move the arrow to Cassette View as shown in Figure 8 and click once to return to the Measure Select Screen as shown in Figure 9. The measurement station in red indicates that this station is currently performing measurement. At this time, select the recipes for vacant measurement wafer stations according to steps A to D (without pressing Start) to continue measuring.

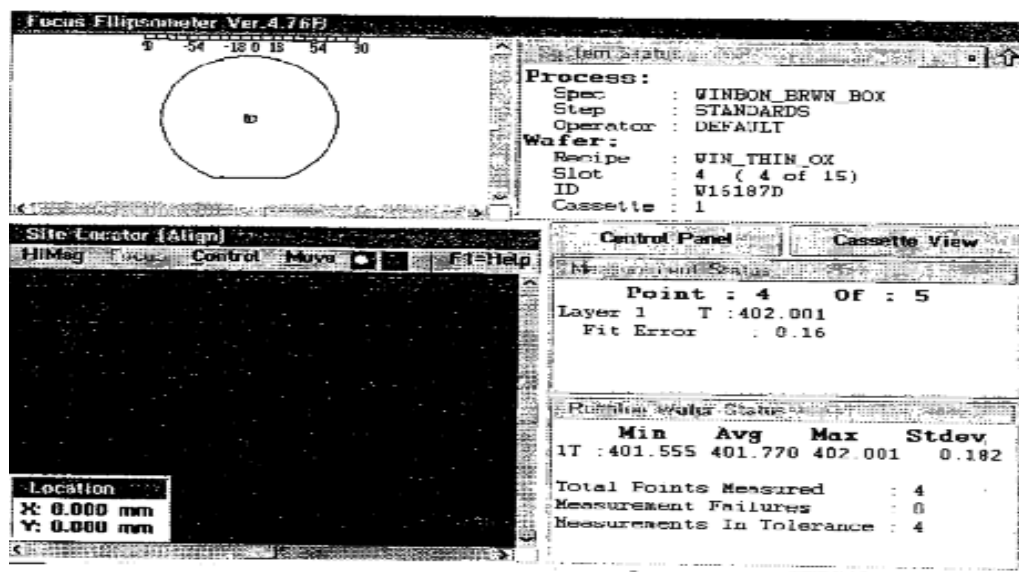


圖 (8)

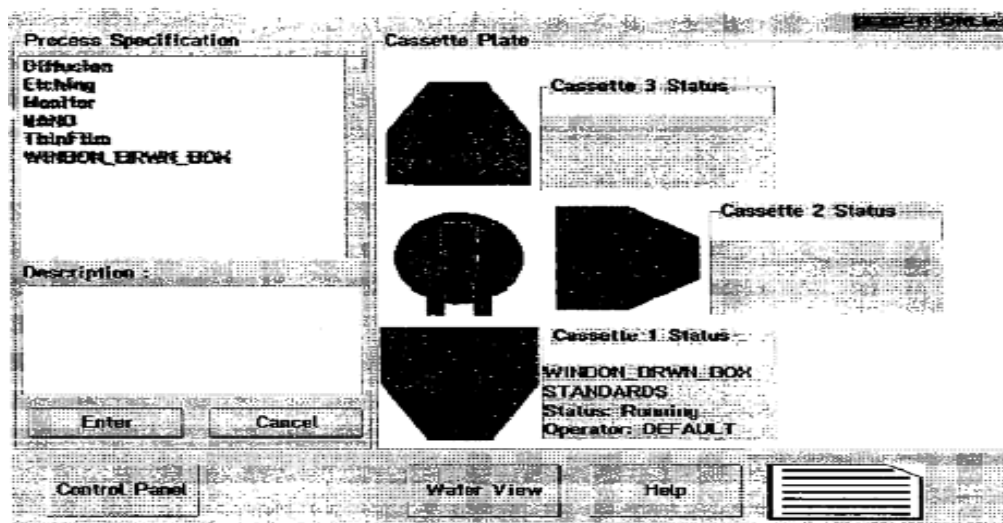


圖 (9)

9. 由圖九回到圖八只要在圖九中移動箭號在 Wafer View 位置按一下即可。

To return from the screen in Figure 9 to Figure, move the arrow to Wafer View and click once.

10. 進行氮化矽產品量測時，在選完程式後會出現輸入前層氧化層厚度之詢問畫面(如圖十)，按 OK 進入厚度值輸入畫面，(如圖十一)。用箭號以畫面上之數字盤輸入前層正確厚度後按 Enter，即回到圖四畫面，再按 Start 便可進行量測。

To measure silicon nitride products, the screen displays a window asking you to enter

to the detected thickness of the previous oxide layer (as shown in Figure 10). Press ok to enter the "Enter Film Value" screen (in Figure 11). Move the arrow to the correct thickness value on the numeric keypad on the screen and press Enter. The screen returns to the screen in Figure 4. Press Start to start measuring.

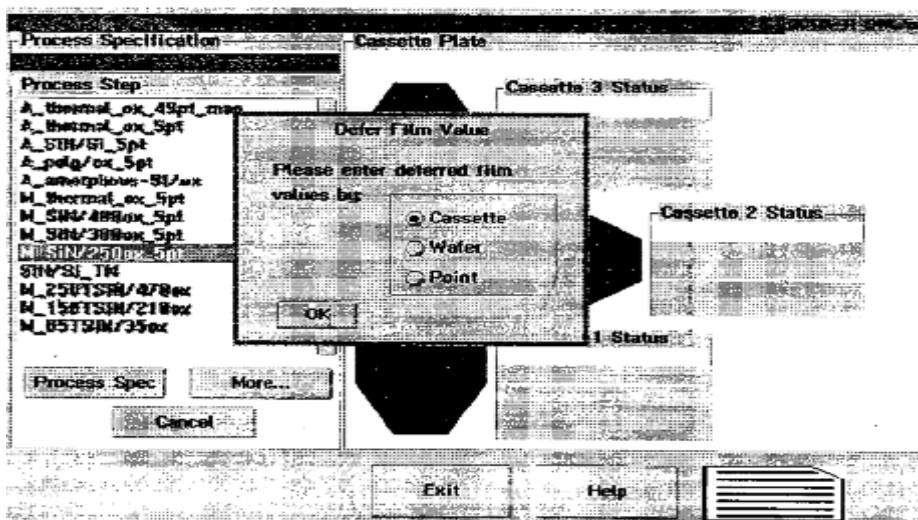


圖 (10)

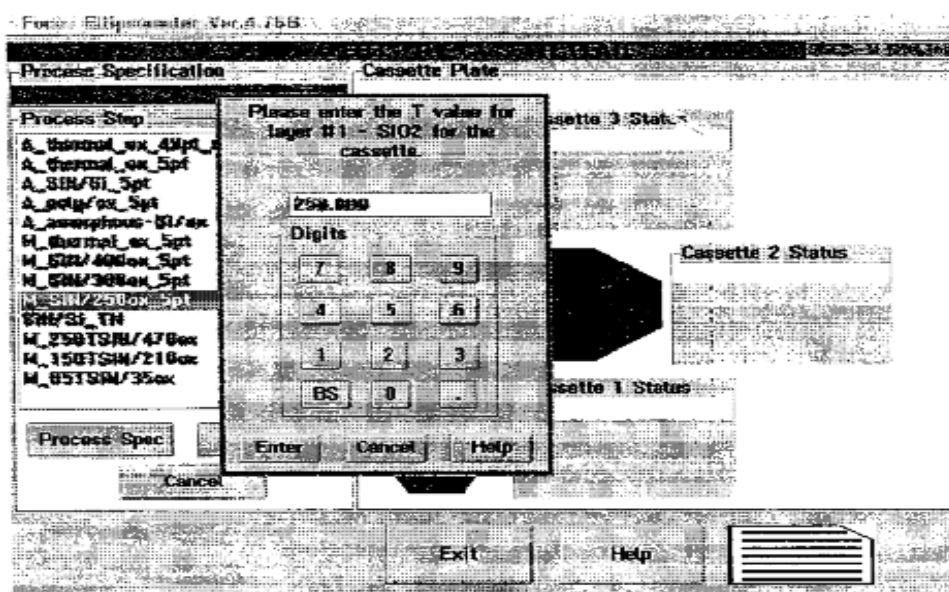


圖 (11)

NO.:	G3214-0904	VERSION:	Y	PAGE:	9
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肆、程式內容(Content of recipe)

- 各製程內之程式命名以 "A" 字母開頭者為自動量測，以 "M" 字母開頭者為手動量測，程式名稱最後以數字加 "pt" 代表該程量測之點數。

Recipe name starting with "A" indicates is auto run while "M" means manual run. The number plus "pt" at the end of the recipe name indicate the number of points measured by this recipe.

- 量測點之座標位置為 (The point measurement coordinates are)

(1) 量測 5 點之座標位置為 (The 5-point measurement coordinates are):

(0,65) (0,0) (0,-60) (-65,0) (65,0)

(2) 量測 9 點之座標位置為(The 9-point measurement coordinates are):

(0,68) (0,0) (0,-62) (-68,0) (68,0) (0,34) (0,-34) (-34,0) (34,0)

(3) 測49點為 Rudolph 標準之 wafer map 量測位置。

The 49-point measurement is the Rudolph standard wafer map measurement positions.

- 量測程式名稱因隨工廠製程之變動而異將隨時以 Memo 之形式公布。

The measurement recipe names change according to the changes in the processes and will be announced through memos.

- 量測程式對照表如下：

FILM	程式名稱 <i>Recipe name</i>	EDC/RTC	適用厚度 <i>Application thickness</i>	適用製程 <i>Application process</i>
OXIDE	A_thermal_OX>500A	RTC	>500A	墊層氧化 (<i>PAD</i>),SAC 氧化 (<i>SAC</i>),井區擴散 (<i>Drive-in</i>),場氧化 (<i>FOX</i>),閘層氧化 (<i>GOX</i>),S/D 氧化,復晶氧化,PLUG 氧化 (<i>Poly oxide</i>),TEOS
	A_thermal_OX<500A	RTC	<500A	
	M_thermal_OX>500A	EDC	>500A	
	M_thermal_OX<500A	EDC	<500A	
LP-POLY	A_Poly<2200A/OX 1000_5pt	RTC	POLY<2200A /TEOS~1000A	LP-POLY
	A_Poly>2200A/OX 1000_5pt	RTC	POLY>2200A ,/TEOS~1000A	

NO.:		G3214-0904		VERSION:	Y	PAGE:	10
	A_Poly<2200A/OX 1500_5pt	RTC	POLY<2200A /TEOS~1500A	LP-POLY			
	A_Poly>2200A/OX 1500_5pt	RTC	POLY>2200A /TEOS~1500A				
	A_Poly<2200A/OX 2000_5pt	RTC	POLY<2200A /TEOS~2000A				
	A_Poly>2200A/OX 2000_5pt	RTC	POLY>2200A /TEOS~2000A				
	M_PROPOLY/OX_5pt	EDC	GOX~120A	LP-POLY for E06xx			
	M_PROPOLY/105 OX_5pt	EDC	GOX~105A	LP-POLY for E05xx			
	M_PROPOLY/90 OX_5pt	EDC	GOX~90A	LP-POLY for E04xx			
LP-SIN	A_SIN/150_OX_5pt	EDC	4217/Pad~150A	4217 EDC			
	A_SIN/250_OX_5pt	EDC	4217/Pad~250A				
	A_SIN/400_OX_5pt	EDC	4217/Pad~400A				
	A_SIN/SI_5pt	RTC	M/W for 5914 or Thin_SIN	5914 and Thin_SIN(90/150/250A)			

5. 針對 EM-5 量測厚度 OXIDE 500~1000A(不含 TEOS1000A) · 請改用以下程式:

- 自動量測 : A_thermal_OX_1000-500A_5pt
- 手動量測 : M_thermal_OX_1000-500A_5pt

EM-5 for oxide thickness 500~1000A(not included TEOS 1000A) · the recipe are:

- Auto measurement : A_thermal_OX_1000-500A_5pt
- Manual measurement : M_thermal_OX_1000-500A_5pt

伍、自動化操作事項 (Automated communication operation items)

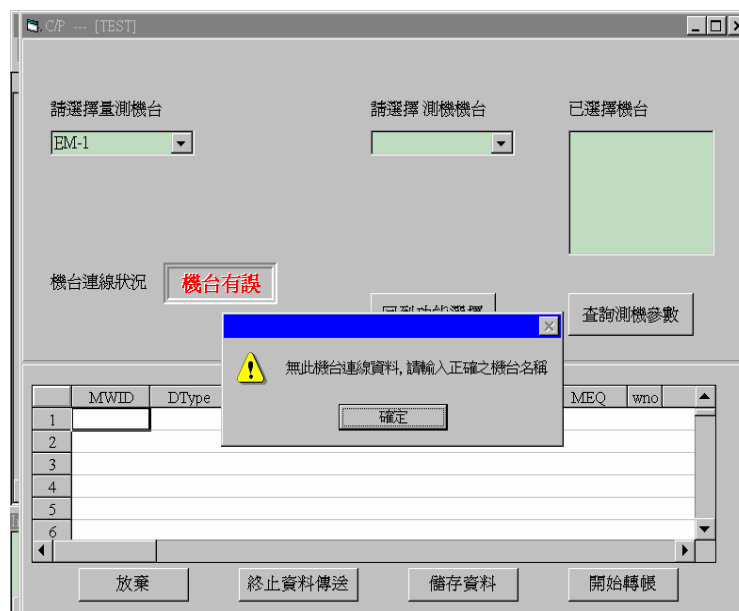
1. 輸入 operator 工號及密碼

Login : to input the user work number password.



2. 若顯示於機台連線狀況異常或不存在, 請重新選擇量測機台。

If occurs communication link error or doesn't exist, please re-select measurement machine.



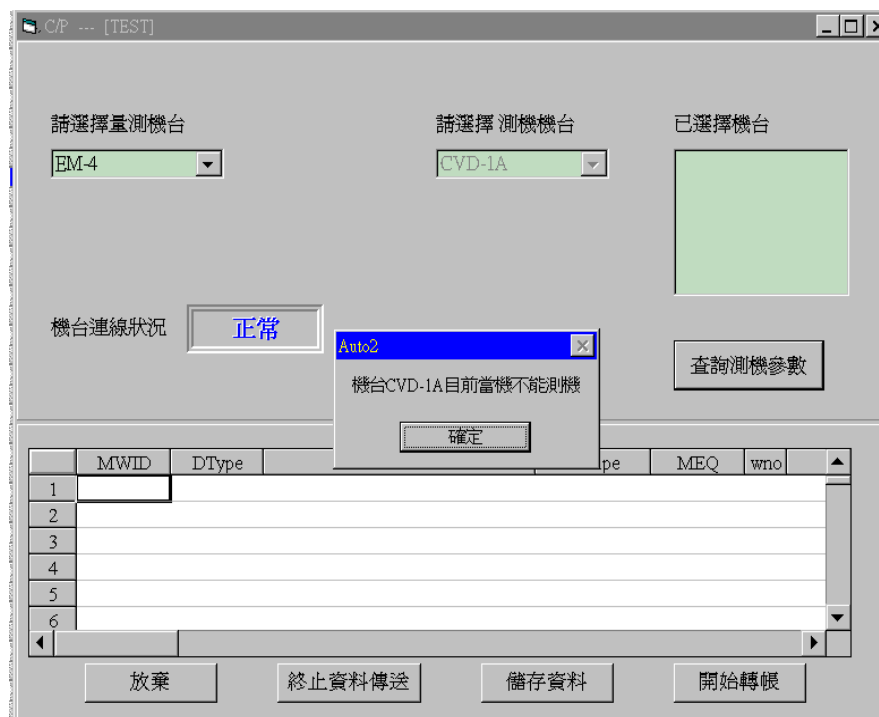
3. 選擇量測機台, 連線狀況正常

This figure show communication correct



4. 選擇測機機台, 若機台狀況不允許作測機, operator 無法選擇此機台

If process machine status is not correct · terminal system will show you error message.



NO.:

G3214-0904

VERSION:

Y

PAGE:

13

5. 可多選測機機台, 之後按查詢測機參數, 開始量測及參數對應工作

You could select multiple c/p machines and check measurement parameter to communication

6. Operator 開始量測

To start to measure

NO.:

G3214-0904

VERSION:

Y

PAGE:

14

7. 量出之 raw data 會即時顯示於下方表格內

Terminal system will show measurement data

The screenshot shows the CVD-2A software interface. It has a title bar 'C/P --- [TEST]' and two tabs: 'CVD-2A' and 'CVD-2B'. The main area contains two tables. The top table lists parameters with columns: parameter name, 參數名稱, 參數值, and 原始資料. The bottom table lists raw data with columns: MWID, DType, Raw Data, Recipe, MEQ, and wno. At the bottom are four buttons: 放棄, 終止資料傳送, 儲存資料, and 開始轉帳.

	parameter name	參數名稱	參數值	原始資料
1	PARTICLE-	PARTICLE		前後
2				
3	OX-E-R	OX-E-R		
4	OX-STRESS	OX-STRESS		
5	OX-THK	OX-THK		
6	OX-THK-U	OX-THK-U		
7	OX-RI	OX-RI		

	MWID	DType	Raw Data	Recipe	MEQ	wno
1		R	5196.17 4790.87 4894.48 4991.1		EM-5	1 5
2		R	5018.93 4928.89 4989.68 5193.5		EM-5	2 5
3		R	5114.41 4932.92 4831.82 4787.4		EM-5	3 5

Buttons: 放棄, 終止資料傳送, 儲存資料, 開始轉帳

8. Operator 將下方量測資料對應至某參數列, 系統自動計算參數值

You select the first measurement data in order to input to correct parameter site

The screenshot shows the CVD-2A software interface with the same layout as the previous one. In this state, the first row of the raw data table (MWID 1) is highlighted in green, indicating it is selected for calculation. The 'OX-THK' parameter row in the top table now has a value of 4965.878 in the '參數值' column, which is the result of the calculation based on the selected raw data.

	parameter name	參數名稱	參數值	原始資料
1	PARTICLE-	PARTICLE		前後
2				
3	OX-E-R	OX-E-R		
4	OX-STRESS	OX-STRESS		
5	OX-THK	OX-THK	4965.878	5196.17 4790.87 48
6	OX-THK-U	OX-THK-U		
7	OX-RI	OX-RI		

	MWID	DType	Raw Data	Recipe	MEQ	wno
1		R	5196.17 4790.87 4894.48 4991.1		EM-5	1 5
2		R	5018.93 4928.89 4989.68 5193.5		EM-5	2 5
3		R	5114.41 4932.92 4831.82 4787.4		EM-5	3 5

Buttons: 放棄, 終止資料傳送, 儲存資料, 開始轉帳

9. 對應另一測機機台同類參數

You select the second measurement data in order to input to next parameter site

The screenshot shows the 'C/P --- [TEST]' window with the 'CVD-2B' tab selected. It contains two tables and several buttons.

	parameter name	參數名稱	參數值	原始資料
1	PARTICLE-	PARTICLE		前
2				後
3	OX-E-R	OX-E-R		
4	OX-STRESS	OX-STRESS		
5	OX-THK	OX-THK	5028.012	5018.93 4928.89 49
6	OX-THK-U	OX-THK-U		
7	OX-RI	OX-RI		

	MWID	DType	Raw Data	Recipe	MEQ	wno
1		R	5196.17 4790.87 4894.48 4991.1	EM-5	1	5
2		R	5018.93 4928.89 4989.68 5193.5	EM-5	2	5
3		R	5114.41 4932.92 4831.82 4787.4	EM-5	3	5

Buttons at the bottom: 放棄, 終止資料傳送, 儲存資料, 開始轉帳

10. Operator 先儲存已對應好的參數值，Operator 開始另一種類之參數量測流程

To save input data，another select data flow (for particle measurement) you select measure-particle-machine as SS-4

The screenshot shows the 'C/P --- [TEST]' window with a different layout. It includes dropdown menus for machine selection and a status indicator.

請選擇量測機台: SS-4

請選擇 測機機台: CVD-2B

已選擇機台: CVD-2A, CVD-2B

機台連線狀況: 正常

Buttons: 回到功能選擇, 查詢測機參數

	MWID	DType	Raw Data	Recipe	MEQ	wno
1						
2						
3						
4						
5						
6						

Buttons at the bottom: 放棄, 終止資料傳送, 儲存資料, 開始轉帳

11. 之前存入的參數值會自動帶出

You could get it that be saved data before

The screenshot shows the CVD-2A software interface. At the top, there are tabs for CVD-2A and CVD-2B. Below the tabs is a table with columns: parameter name, 參數名稱, 參數值, and 原始資料. The table contains data for parameters like PARTICLE, OX-E-R, OX-STRESS, OX-THK, OX-THK-U, and OX-RI. Below this table is another table with columns: MWID, DType, Raw Data, Recipe, MEQ, and wno. A large blue message box is overlaid on the bottom table, containing the text: 等待儀器量測資料中, 請稍候. At the bottom of the interface are four buttons: 放棄, 終止資料傳送, 儲存資料, and 開始轉帳.

12. 進行其他參數值對應

To combine particle parameter valve

The screenshot shows the CVD-2A software interface. At the top, there are tabs for CVD-2A and CVD-2B. Below the tabs is a table with columns: parameter name, 參數名稱, 參數值, and 原始資料. The table contains data for parameters like PARTICLE, OX-E-R, OX-STRESS, OX-THK, OX-THK-U, and OX-RI. Below this table is another table with columns: MWID, DType, Raw Data, Recipe, MEQ, and wno. This table has four rows highlighted in green. At the bottom of the interface are four buttons: 放棄, 終止資料傳送, 儲存資料, and 開始轉帳.

NO.:

G3214-0904

VERSION:

Y

PAGE:

17

13. 所有此次測機所須的參數都對應完成後開始進行轉帳

To get all necessary parameter valve in order to input and move out

	parameter name	參數名稱	參數值	原始資料
1	PARTICLE-	PARTICLE	1	前 0
2				後 1
3	OX-E-R	OX-E-R		
4	OX-STRESS	OX-STRESS		
5	OX-THK	OX-THK	4965.878	4965.878
6	OX-THK-U	OX-THK-U		
7	OX-RI	OX-RI		

	MWID	DType	Raw Data	Recipe	MEQ	wno
1		P	0		SS-3	1 1
2		P	4		SS-3	2 1
3		P	1		SS-3	3 1
4		P	2		SS-3	4 1

放棄 終止資料傳送 儲存資料 開始轉帳

14. 轉帳畫面

Display move out function

	parameter name	參數名稱	參數值	原始資料
1	PARTICLE-	PARTICLE	-2	前 4
2				後 2
3	OX-E-R	OX-E-R		
4	OX-STRESS	OX-STRESS		
5	OX-THK	OX-THK	5028.012	5028.012
6	OX-THK-U	OX-THK-U		
7	OX-RI	OX-RI		

☒ CVD-2A
☒ CVD-2B

開始進行轉帳
 回到參數對應
 離開測機流程

NO.:	G3214-0904	VERSION:	Y	PAGE:	18
------	------------	----------	---	-------	----

陸、操作注意事項 (Operation notes)

1. 嚴禁震動或搖晃機台。

It is strictly prohibited to vibrate or rock the machine.

2. 機台前門禁止開啟以免造成傳送動作異常而當機。

Do not open the front machine door to prevent the machine from hanging due to transfer error.

3. 嚴禁離開至本 O.I.所規定以外之畫面，如遇軟體問題而跳離，請勿任意動作並儘速洽EQ處理。

Do not exit to screens not regulated in this O.I. If exit is due to software problem, do not take any action but consult the engineer as quickly as possible for processing.

4. 晶片卡在機台內應通知工程師處理，嚴禁擅自強行取出晶片。

If wafer is jammed inside the machine, consult with PE. Do not try to free the wafer by force.

5. 嚴禁擅自關閉及開啟電源或按RESET鍵，以免造成軟、硬體上之損壞，若有問題PE或EQ處理

It is strictly prohibited to turn the power on or off, or press the RESET key to prevent damaging the software or hardware. Consult the PE or EQ for processing for any machine problem.

6. EM-5手動量測第一點如果是 C-key 位置，請移到 U-key 位置，再開始量測。

Please move to U-key and start if EM-5 manual measure first point is C-key position.

柒、例行作業(Routine operation)

1. EA 每週三 AM 08:00 ~12:00 由日班使用自製測機控片執行測機。

Every Wednesday, EA must do calibration (day shift only).

2. 以控片分別量測 EM-4、EM-5、EM-6 機台之各種層次厚度，量測值使用 auto -C/P轉帳。

Use the monitor wafer to measure the thicknesses of all layers in the EM-4,EM-5,EM-6 machines, then use auto-C/P to transfer the data.

NO.:	G3214-0904	VERSION:	Y	PAGE:	19
------	------------	----------	---	-------	----

3. 控片量測前將 wafer 平邊朝下再進行量測。

Wafer of flat transfer to underside at monitor wafer measure before.

捌、品質控制 (Quality control)

1. 標準晶片須每年由儀校管理單位 委外校驗，每半年使用標準晶片校正 EM-4、EM-5、

EM-6 機台再使用 Nitride 1500A 及 Poly 1500A 自製測機控片做 GR&R 之量測驗證。

Every year, must take Rudolph standard wafer to QC to do wafer calibrate, then every half year do GR&R calibration.

2. 標準晶片分為二種 (Standard Wafer)

2.1 Rudolph 標準片：厚度為 oxide 100A、400A、1100A、2300A 及 Nitride 950A。

2.2 自製標準控片：厚度為 oxide 70A、oxide 10000A。

3.3 標準片之規格及量測點則依標準校驗文件(certification sheet)規定。

From vendor to get calibrate wafer, the thickness includes oxide 100A, 400A,

1100A, 2300A and Nitride 950A. The correct specification and measurement points of the Standard Wafer are as regulated in the Certification Sheet.

3. Rudolph 標準片規格、機台規格與自製標準控片規格如下表所示

Standard wafer includes two kinds Rudolph std wafer and self made std wafer:

標準片規格 (golden wafer spec) :

請上“ 儀器校驗報告書電子表單系統 http://nthcweb01/CMS/CMS_EFORM/ ”查詢。

CMS E-Form System “http://nthcweb01/CMS/CMS_EFORM/”

(範例如下) **Example as below**

詳細規格
項次：1
校驗項目：Sio2 厚度
儀器使用(點)範圍：93A(固定折射率 1.462)
允收規格：± 2 A
詳細規格
項次：1
校驗項目：Sio2 厚度
儀器使用(點)範圍：10175A(固定折射率 1.462)
允收規格：± 25 A

NO.:	G3214-0904	VERSION:	Y	PAGE:	20
------	------------	----------	---	-------	----

標準片	標準片規格
OXIDE 70 (自製)	外校量測值與厚度標準值差異需在 小於或等於 ± 2 內
OXIDE 100 (RUDOLPH)	外校量測值與厚度標準值差異需在 小於或等於 ± 2 內
OXIDE 400 (RUDOLPH)	外校量測值與厚度標準值差異需在 小於或等於 ± 5 內
OXIDE 1100 (RUDOLPH)	外校量測值與厚度標準值差異需在 小於或等於 ± 6 內
OXIDE 2300 (RUDOLPH)	外校量測值與厚度標準值差異需在 小於或等於 ± 11 內
OXIDE 10000 (自製)	外校量測值與厚度標準值差異需在 小於或等於 ± 25 內
NITRIDE 950 (RUDOLPH)	外校量測值與厚度標準值差異需在 小於或等於 ± 6 內

標準片保存條件 (golden wafer storage condition)

標準片日常保管於恆溫恆濕之潔淨室內,同晶圓廠條件。標準晶片須每年由 儀校管理部委外校驗，若校驗值超過 $\pm 1.5A$ 則做 re-target

Golden wafers are stored in a clean room with constant temperature and humidity, which is the same as that of the wafer fab. Golden wafers must be managed External verification by QC every year. If the verification value exceeds $\pm 1.5A$, re-target

厚度標準值請參考儀校經歷卡. *Thickness target search from CMS E-Form System*

Rudolph 標準片for 機台規格 (ellipsometer SPEC for Rudolph standard wafer)

請上” 儀器校驗報告書電子表單系統 http://nthcweb01/CMS/CMS_EFORM/ “ 查詢。

CMS E-Form System “http://nthcweb01/CMS/CMS_EFORM/”

(範例如下) *Example as below*

NO.:	G3214-0904	VERSION:	Y	PAGE:	21
------	------------	----------	---	-------	----

項次：1 校驗項目：OXIDE-73					
儀器使用(點)範圍：73A					
允收規格：± 4 A					
規格計算方式：B2-A					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1	73	NA		A	
項次：2 校驗項目：OXIDE-100					
儀器使用(點)範圍：93A					
允收規格：± 4 A					
規格計算方式：B2-A					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1	93	NA		A	
項次：3 校驗項目：OXIDE-400					
儀器使用(點)範圍：389A					
允收規格：± 10 A					
規格計算方式：B2-A					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1	389	NA		A	
項次：4 校驗項目：OXIDE-1100					
儀器使用(點)範圍：1137A					
允收規格：± 25 A					
規格計算方式：B2-A					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1	1137	NA		A	
項次：5 校驗項目：OXIDE-2300					
儀器使用(點)範圍：2322A					
允收規格：± 45 A					
規格計算方式：B2-A					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1	2322	NA		A	
項次：6 校驗項目：NITRIDE -950					
儀器使用(點)範圍：953A					
允收規格：± 25 A					
規格計算方式：B2-A					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1	953	NA		A	
項次：7 校驗項目：OXIDE-10000					
儀器使用(點)範圍：10175A					
允收規格：± 100 A					
規格計算方式：B2-A					
測量(點)範圍					
item	標準件-讀值(A)	被校件-設定(B1)	被校件-讀值(B2)	單位	備註
1	10175	NA		A	

標準片	Ellipsometer SPEC for standard wafer
OXIDE 70 (自製)	與厚度標準值差異需在小於或等於 +/-4內
OXIDE 100 (RUDOLPH)	與厚度標準值差異需在小於或等於 +/-4內
OXIDE 400 (RUDOLPH)	與厚度標準值差異需在小於或等於 +/-10內
OXIDE 1100 (RUDOLPH)	與厚度標準值差異需在小於或等於 +/-25內
OXIDE 2300 (RUDOLPH)	與厚度標準值差異需在小於或等於 +/-45內

NO.:	G3214-0904	VERSION:	Y	PAGE:	22
------	------------	----------	---	-------	----

OXIDE 10000 (自製)	與厚度標準值差異需在小於或等於+/-100內
NITRIDE 950 (RUDOLPH)	與厚度標準值差異需在小於或等於+/-25內

EM機台每半年使用標準晶片做校驗，若校驗值超出上述規格，則EM機台需校正上述規格內，厚度標準值請參考儀校經歷卡。

The EM machine uses Golden wafer for calibration every 6 months.

If the calibration value exceeds the above specifications, the EM machine needs to be calibrated within the above specifications.

Thickness target search from CMS E-Form System

- 自製測機控片規格 (Self made MW wafer spec) · 用途：EA 例行作業量測 EM 機台使用的控片。

EM –項目 (ITEM)	規格 (SPEC)
Slot 01_thermal ox 7xA	SPEC= 依每年再確認目標值 $\pm 2A$
Slot 02_thermal ox 10xA	SPEC= 依每年再確認目標值 $\pm 4A$
Slot 03_thermal ox 17xA	SPEC= 依每年再確認目標值 $\pm 4A$
Slot 04_thermal ox 35xA	SPEC= 依每年再確認目標值 $\pm 5A$
Slot 05_thermal ox 43xA	SPEC= 依每年再確認目標值 $\pm 5A$
Slot 06_thermal ox 88xA	SPEC= 依每年再確認目標值 $\pm 7A$
Slot 07_thermal ox 10xxA	SPEC= 依每年再確認目標值 $\pm 7A$
Slot 08_thermal ox 16xxA	SPEC= 依每年再確認目標值 $\pm 7A$
Slot 09_thermal ox 19xxA	SPEC= 依每年再確認目標值 $\pm 10A$
Slot 10_thermal ox 24xxA	SPEC= 依每年再確認目標值 $\pm 12A$
Slot 11_thermal ox 64xxA	SPEC= 依每年再確認目標值 $\pm 20A$
Slot 12_thermal ox 10xxxA	SPEC= 依每年再確認目標值 $\pm 28A$
Slot 13_poly/ox 54xA	SPEC= 依每年再確認目標值 $\pm 4A$
Slot 14_poly/ox 14xxA	SPEC= 依每年再確認目標值 $\pm 7A$
Slot 15_poly/ox 37xxA	SPEC= 依每年再確認目標值 $\pm 20A$

NO.:	G3214-0904	VERSION:	Y	PAGE:	23
------	------------	----------	---	-------	----

Slot 16_Nitride 9xA	SPEC= 依每年再確認目標值 $\pm 2A$
Slot 17_Nitride 14xxA	SPEC= 依每年再確認目標值 $\pm 5A$

NANO –項目 (ITEM)	規格 (SPEC)
Slot 01_thermal ox 7xA	SPEC= 依每年再確認目標值 $\pm 10A$
Slot 02_thermal ox 10xA	SPEC= 依每年再確認目標值 $\pm 10A$
Slot 03_thermal ox 17xA	SPEC= 依每年再確認目標值 $\pm 10A$
Slot 04_thermal ox 35xA	SPEC= 依每年再確認目標值 $\pm 15A$
Slot 05_thermal ox 43xA	SPEC= 依每年再確認目標值 $\pm 15A$
Slot 08_thermal ox 16xxA	SPEC= 依每年再確認目標值 $\pm 7A$
Slot 10_thermal ox 24xxA	SPEC= 依每年再確認目標值 $\pm 12A$
Slot 11_thermal ox 64xxA	SPEC= 依每年再確認目標值 $\pm 20A$
Slot 12_thermal ox 10xxxA	SPEC= 依每年再確認目標值 $\pm 28A$
Slot 13_poly/ox 54xA	SPEC= 依每年再確認目標值 $\pm 4A$
Slot 15_poly/ox 37xxA	SPEC= 依每年再確認目標值 $\pm 20A$
Slot 17_Nitride 14xxA	SPEC= 依每年再確認目標值 $\pm 5A$

5. 自製測機控片之製做程序 (The procedure to make self-made MW wafer) :

5.1. 自製測機控片可持續使用無保存年限但若有遺失，損毀則須重新制做。

If standard wafer is lost or damaged , please to re-produce the new one.

5.2. 選用全新控片後經刻號，清洗後，置於條件穩定之爐管內，長出所須求之膜與厚度，自製測機控片之刻號內容須標明膜厚與膜之種類。

NO.:	G3214-0904	VERSION:	Y	PAGE:	24
------	------------	----------	---	-------	----

Please marking a new wafer · cleaning then grow the desired film thickness. The self made standard wafer have to mark the sign to identity properly and film thickness.

- 5.3. 因所長出之薄膜，初期其膜厚仍尚未穩定，須每日於量測機台 (EM-4、5、6) 量測其厚度對時間之變化，待其穩定後，使可開使收集於各機台之量測值，並於收集滿 25 筆 DATA 後依 SPC 管制方法定出規格。

Because the new growth film thickness unstable · we have to measure the film thickness in EM-4 · 5 · 6 every day. After film thickness is stable · please measure 25 data to define the spec by the SPC rules.

- 5.4. 製做完成測機控片後，另行知會 S500 更改 NANO 8000 之規格。

After finished · please inform to S500 to change the spec of NANO 8000 machine.

- 5.5. 自製測機控片每半年做一次目標值的再確認，並將量測值記錄於網站上。

Self made standard wafer every half year do GR&R calibration.

- 5.6. 自製測機控片平常時置於恆溫恆濕之潔淨室內，每半年 EM 機台校正回來後做自製測機控片目標值再確認的量測，若其量測值與原目標值差異大於一個 σ ，則將目標值更改為再確認的量測值，若 $\leq 1\sigma$ 則不做更動。

Standard wafer have to be stored in clean room for constant temperature and humidity. If wafer is damaged · broken or thickness calibration fail · must to produce the new one wafer.

- 5.7. 自製測機控片製做完畢或每半年再確認後則由保管人於 Cassette 外標示出規格與 Slot 位置以資識別且需記錄更新日期與有效日期以及需有 S210 & S510 PE 蓋章方有效，範例如下圖。

NO.:	G3214-0904	VERSION:	Y	PAGE:	25
------	------------	----------	---	-------	----

FILM	Light	EM /NANO (自製控片)	Slot	Ellipsometer(EM) OOS spec			NANO OOS spec		
				LSL	Target	USL	LSL	Target	USL
Thin oxide	117 UV	Oxide 73A	1	71	73	75	63	73	83
		Oxide 108A	2	104	108	112	98	108	118
		Oxide 176A	3	172	176	180	166	176	186
Thick oxide	101 VIS	Oxide 358A	4	353	358	363	343	358	373
		Oxide 437A	5	432	437	442	422	437	452
		Oxide 886A	6	879	886	893			
		Oxide 1019A	7	1012	1019	1026			
		Oxide 1600A	8	1593	1600	1607	1593	1600	1607
		Oxide 1989A	9	1979	1989	1999			
		Oxide 2496A	10	2484	2496	2508	2484	2496	2508
		Oxide 6499A	11	6479	6499	6519	6479	6499	6519
		Oxide 10186A	12	10158	10186	10214	10158	10186	10214
Poly	104 VIS	Poly 542A	13	538	542	546	538	542	546
	104 VIS	Poly 1493A	14	1486	1493	1500	NA		
	104 VIS	Poly 3735A	15	3715	3735	3755	3715	3735	3755
Nitride	103	Nitride 90A	16	88	90	92	NA		
		Nitride 1435A	17	1430	1435	1440	1430	1435	1440

以系統為主 更新日期:

有效日期:

S210 EM 製程:
S510 NM 製程:

After new standard is made , we have label the thickness spec and slot number in the cassette to identify and renew date, valid period, S210 & S510 PE signature.

- 5.8. 定期儀器之校正時機依儀器經歷卡上之校正週期執行校正，量測標準片比對量測值與標準值並記錄於校正報告上，依經歷卡上的容許誤差值進行是否校正合格之判定，並“0000-1000 儀器校驗管理程序”執行相關校驗規定。

The calibration timing of the regular instrument should be performed according to the calibration cycle on the instrument history card. Measure the standard sheet and compare the measured value with the standard. The accurate value is recorded on the calibration report, and whether the calibration is qualified is determined based on the allowable error value on the experience card, and "0000-1000 Instrument Calibration Management Procedure" implements relevant calibration regulations.

NO.:	G3214-0904	VERSION:	Y	PAGE:	26
------	------------	----------	---	-------	----

玖、儀器 MSA 檢定 (The MSA calibration):

本儀器納入量具 MSA 檢定系統，依量具 MSA 經歷卡之檢定週期進行檢定，檢定之儀器操作同儀器之操作程序。

The instrument is incorporated into the measuring tool MSA calibration system.

It is calibrated according to the calibration cycle of the measuring tool MSA experience card. The calibrated instrument operation procedures for the same instrument

1. GR&R

儀器定期校驗合格後，取合乎量具 MSA 經歷卡上之 wafer，從 EA 輪流各擇取三人，依經歷卡上 GRR 各項規定內容進行檢定

After the instrument qualified the regular calibration, the wafers on the MSA experience card of the qualified measuring instrument will be selected, and three people from each EA will be selected in turn. Conduct the test according to the GRR provisions on the experience card

2. Bias/Linearity

依量具 MSA 經歷卡上所規定之校驗合格標準片，連續進行量測後，進行數據的分析與檢定。

According to the calibration standard sheet specified on the MSA experience card of the measuring tool, after continuous measurement, the data is analyzed and verified.

3. Stability

取量具 MSA 經歷卡上所指定之 WAFER，依其規定內容進行檢定與作管制圖。

Take the WAFER specified on the MSA experience card of the measuring tool and perform verification and make control drawings according to its specified content.

4. MSA 相關作業依規範“4040-0114 MSA (量測系統分析) 作業規範”執行。

5. MSA 檢定規格 (MSA Test Spec)：

請上“儀器校驗報告書電子表單系統 http://nthcweb01/CMS/CMS_EFORM/”查詢。
(範例如下)

Search from “CMS E-Form System http://nthcweb01/CMS/CMS_EFORM/”。

NO.:	G3214-0904	VERSION:	Y	PAGE:	27
------	------------	----------	---	-------	----

%GRR詳細規格								
檢定項目	樣本數	(晶片) 測試點	重複次數	作業員人數	製程規格			
Nitride	5	5	3	3	0.1A	1400	1600	A
Poly	5	5	3	3	0.1A	1340	1640	A
%BIAS詳細規格								
檢定項目				製程規格				
			標準片值	最小刻度	規格下限	規格上限	單位	
OXIDE THICKNESS			73	0.01A	63	77	A	
			93	0.01A	83	97	A	
			389	0.01A	360	440	A	
			1137	0.1A	900	1100	A	
			2322	0.1A	1800	2200	A	
			10175	0.1A	9000	11000	A	
Nitride Thickness			953	0.1A	1400	1600	A	
%Linearity詳細規格								
檢定項目				測量範圍	標準片值		單位	
Oxide Thickness				70~500	73		A	
					93			
					389			
Oxide Thickness				501~10000	1137		A	
					2322			
					10175			
Stability詳細規格								
檢定項目	參數名稱			管制圖形式		檢定頻率		
Oxide Thickness	TOX_108			X-Bar R Chart		Weekly		
Oxide Thickness	TOX_1601			X-Bar R Chart		Weekly		
Oxide Thickness	TOX_6499			X-Bar R Chart		Weekly		

拾、機台 Automate communication 之設定

(Machine Automated Communication Setting)

1. FE-III 機台之 SECS-III 的相關設定：

FE-III machine SECS setting. The following shows the SECS-III relevant settings:

- 離開 FOCUS OPERATOR 模式，輸入 User-ID 及 password。
Exit FOCUS OPERATOR mode and enter User-ID and password.
- 將游標移到 SECS Icon 上按下，然後進入 SECS-II 畫面。
Move the cursor to the SECS Icon and click to enter the SECS-II menu.
- 將 SECS-II Control 的 Current Settings 欄位設為 Local，Power Up Settings 欄位設為 On-Line；並將 SECS-II Communications 的 Current Settings 欄位設為 Enabled，Power Up Settings 欄位設為 Enabled，SECS-II 之畫面，進入此畫面後在 GO OFF-Line 按 4 下再回到 GO OFF-Line)。

Set the SECS-II Control Current Settings field as Local and the Power Up Settings

NO.:	G3214-0904	VERSION:	Y	PAGE:	28
------	------------	----------	---	-------	----

field as On-Line. Also set the SECS-II Communications Current Settings field as Enabled and the Power Up Settings field as Enabled. The SECS-II screen · press 4 in GO OFF-Line to return to GO OFF-Line.

- 當設定完畢後，便能於視窗下方的狀態欄中看到機台和 Cell Controller 之間的通訊。
After setting is complete, you can view the communication between the machine and the Cell Controller in the status field under the window.
- 離開 SECS-II 畫面時，請按左上角將畫面最小化，切記!!不可按右上角功能，否則會將 SECS-II 驅動軟體關閉。

When exiting the SECS-II screen, press the upper left corner to minimize the screen.attention! not to press the function on the upper right corner or it may close the SECS-II driver.

- OS/2 系統的 Startup 視窗內必須有 SECS Icon，否則機台重新開機後，SECS Driver 無法自動執行。此動作僅 PE 或 EQ 可為之。

The OS/2 system Startup window must show the SECS Icon. Otherwise, the SECS driver cannot run when the system reboots. Only the PE or EQ can perform this action.

拾壹、生效與修訂 (*Documentation effectiveness and revision*)

本規範之公佈實施及其修訂核准層級皆依會簽/核決/分發依循範例為之。

The specification shall be announced, implemented, revised and approved in accordance with the levels in the countersignature/approval/distribution examples.